

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. Applied Mathematics & Computer Science

Newark, NJ

Expected: May 2027

Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning

EXPERIENCE

AI/ML Intern, *Apple*

May 2026 – Aug 2026 | Cupertino, CA

- First intern on Agentic Commerce Siri, using Apple Pay and Wallet APIs to execute multi-step agentic workflows with delegated recursive sub-agents.
- Implemented **SpeechAnalyzer** to extract user intent and an **on-device SLM** to generate context-aware instruction.
- Built a **multimodal synthesizer** fusing speech and **On-Screen Awareness** signals to predict user problems and intent.

Machine Learning Intern, *Angeles Investors*

Jan 2026 – May 2026 | New York City, NY

- Built an AI startup diligence agent swarm powered by GPT-4o, Tavily API, Lightpanda browser control.
- Deployed an **asynchronous FastAPI** data ingestion pipeline via **Docker** that **parallelized research agents** and streamed structured results in real time, accelerating startup screening and enrichment.
- Production agent led a **\$500K pre-seed deal** at a \$48M pre-money valuation.

President, *Society of Hispanic Professional Engineers (SHPE)*

Sep 2025 – May 2026 | Newark, NJ

- Led 13 officers fundraising **\$40,000** across **90+** events year-round, growing membership to a record **250+**.
- Launched "internSHPE" a yearly **mentorship program** pairing 3 underclassmen with the presidential team.
- Architected and shipped the chapter's **first React Native app** in **27 days** leading 3 engineers, scaling to **200+ users** in its **first month** with event registration, member details, and mentorship-program features.

AI/ML Intern, *Apple*

May 2025 – Aug 2025 | Cupertino, CA

- Deployed a **LangGraph** agent integrating **RAG** with **MCP** to reason over technical docs for Apple Pay Merchants.
- Used **Qdrant** for **multilingual similarity search** and **GraphQL** to fetch and manipulate data across internal systems.
- Built **LLM evaluation pipeline** benchmarking agent trajectories, tool-use success, groundedness, and hallucination rate.

Software Engineering Intern, *Caterpillar Inc.*

May 2024 – Aug 2024 | Chicago, IL

- Implemented data validation, transformation, and storage in **PostgreSQL** for Autonomous Vehicle Fleet Management via **Apache Airflow** with a **CI/CD** pipeline.
- Fine-tuned transformer models reaching 89% classification accuracy for predictive maintenance of heavy machinery.
- Served end-to-end data pipelines via **FastAPI** microservices at **sub-100ms** latency for 10K+ daily requests.

PROJECTS

Evolv | [Devpost](#)

Flask, NumPy, Pandas, Python

- Placed **1st of 546** teams at HackPrinceton, building an entire backend and predictive modeling system in under 12 hours.
- Built a **geospatial simulation** platform using Census, EIA, and weather APIs to quantify data-center siting tradeoffs.
- Productionized a **real-time Kubernetes pipeline** with LLM-generated reports, data ingestion into **PostgreSQL** and cutting end-to-end analysis time to under 5 seconds.

Founder - HAI

Transformers, PyTorch, pvector, Redis, Docker, FastAPI, LangGraph, Python

- Commercialized an AI-powered SAT prep platform, selling to NJ tutoring centers supporting **600 students** concurrently.
- Built **multimodal** input/output pipelines for PDFs, images, and text, combining **OCR**, vector search, semantic retrieval, prompt orchestration, and **GPT-4o API** to produce personalized questions, explanations, and study material.
- Implemented a **symbolic verifier** using programmatic solvers, constraint checks, structured outputs, and deterministic validation enforcing **100%** solution correctness before delivery.

NeuroCache: KV-Cache Memory Manager | [GitHub](#)

PyTorch, ONNX, llama.cpp, Transformers.js, C++

- Implemented a KV-cache branching for **llama.cpp** with attention-scored eviction and slot-based allocation.
- Designed **cache-replacement algorithms** balancing memory footprint against response quality for long-context sessions.
- Benchmarked eviction strategies across context lengths, quantifying memory-efficiency and generation-quality tradeoffs.

SKILLS

Languages: Python, C++, JavaScript/TypeScript, Swift

AI/ML: PyTorch, scikit-learn, Hugging Face Transformers, ONNX, LangGraph, RAG, MCP, Qdrant, llama.cpp

Systems & Infrastructure: Docker, Git, Kubernetes, PostgreSQL, Redis, REST, GraphQL, FastAPI, Apache Airflow, CI/CD, Flask, NumPy, Pandas, React